

Optical Navigation Investigation Using Open-Source NASA Goddard Software Tool

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This report presents an overview of how a 2D camera images can be utilized to assist in 3D pose estimation. The necessary process of camera calibration is discussed, including the definition of the pinhole camera model, its associated coordinate frames, and the Brown-Conrady light distortion model. After establishing a foundation in optical navigation (OpNav), the Goddard Imaging Analysis and Navigation Tool (GIANT) [2] is used to calculate calibration parameters and conduct optical navigation using images taken from NASA's DAWN mission to the asteroid Vesta. This work demonstrates the capability to use GIANT without requiring an in-depth understanding of the underlying mathematics.

I. Introduction

One can rationally assume that there is a correlation between a 2D image and 3D space. Mathematically defining this relationship is difficult and has been a scientific pursuit since the early twentieth century, beginning with the mathematical analysis of light refraction [5]. Today, using cameras for optical navigation is of extreme relevance to beyond-Earth spacecraft missions where the global navigation satellite system constellations are not available.

An OpNav background is provided to describe the process of utilizing a 2D image in 3D pose estimation. The pinhole camera model is described and a brief discussion of the Brown-Conrady light distortion model is given. Once familiar with OpNav, an overview of GIANT's capabilities are provided. Camera calibration and OpNav results are shown using images from the NASA's DAWN mission to the asteroid Vesta. This work highlights the ability to utilize GIANT without understanding complex OpNav mathematics.

II. Optical Navigation Background

This section will discuss the processes necessary to correlated a 2D image to 3D space. Before an image can be of any navigational utility, a processes known as camera calibration must first take place. Camera calibration is the process of estimating mathematically modeled calibration parameters that ensure accurate 2D image to 3D space transformation.

The mathematical relationship between a 2D image and 3D space includes a camera model and distortion model. In Section II.A the pinhole camera model is discussed and in Section II.B the Brown-Conrady light distortion model is introduced.

A. Pinhole Camera Model

This section aims to provide an introduction to the pinhole camera model and relies on the information provided in [1]. The pinhole camera model models the camera pupil as an infinitesimally small pinhole and includes five reference frames: the body frame (B), the camera frame (C), the focal plane frame (F), the pixel frame (P) and image plane (I). A diagram of the pinhole camera models reference frames is shown in Fig. 1. Through a chaining of rotations the goal is to accurately estimate a pixel location in the P frame that corresponds to a given a 3D point in the B frame, this is known as the "forward mapping problem". Estimating a 3D location in the B frame that corresponds to a given a 2D point in the P frame, this is known as the "reverse mapping problem".

If given the position and orientation of the C frame, as well the as position of the B frame with respect to (w.r.t.) a common reference frame and the camera focal length. The chaining process of reference frames to arrive at a pixel location in the P frame is possible. This process is not accurate due to the distortion of light, which must also be mathematically modeled by a distortion model to ensure an accurate 2D pixel to / from 3D space transformation.

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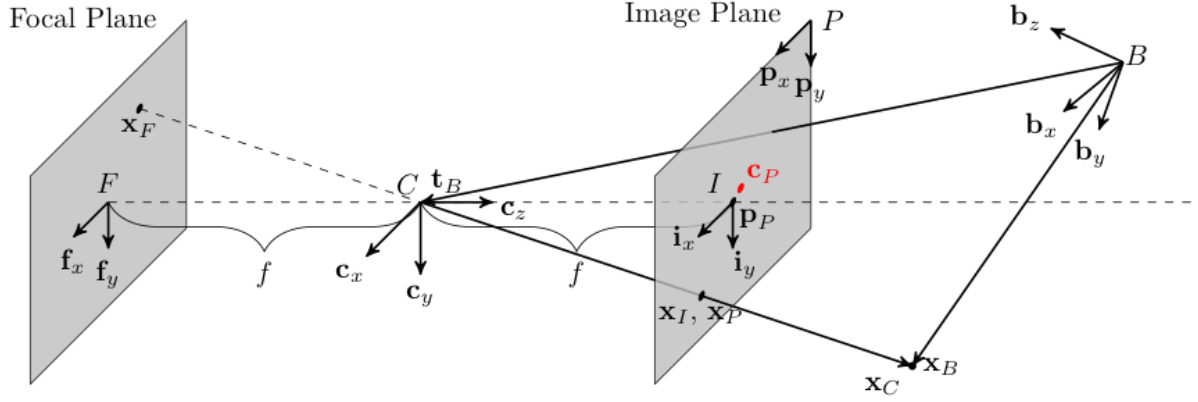
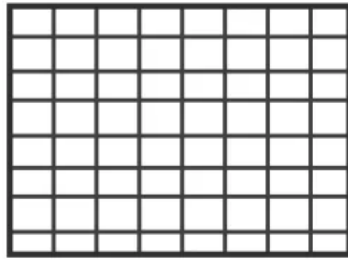


Fig. 1 Pinhole Camera Model Diagram (Referenced from [3])

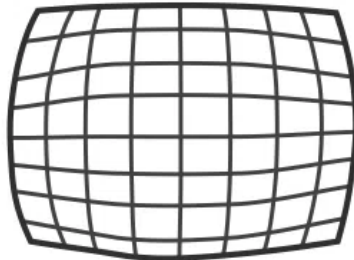
B. Brown-Conrady Distortion Model

Lens distortion can be categorized by tangential and radial distortion, depicted in Fig. 2. Given a set of ideal image plane coordinates, (x, y) , the Brown-Conrady distortion model maps these ideal coordinates to a distorted location, (x_d, y_d) . The Brown-Conrady distortion model uses the parameters, (k_1, k_2, k_3) , to model radial distortion and (p_1, p_2) to model tangential distortion. r is the radial distance of the point of interest from the optical center of the image. The Brown-Conrady distortion model is given by Eq. 1 where $(\cdot)_r$ represents radial distortion and $(\cdot)_t$ tangential distortion.

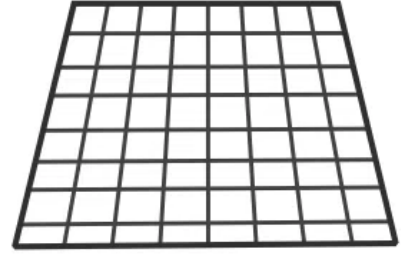
$$\begin{aligned} x_d &= x_r + x_t = x(1 + k_1 r^2 + k_2 r^4 + k_3 r^6) + 2p_1 xy + p_2(r^2 + 2x^2) \\ y_d &= y_r + y_t = y(1 + k_1 r^2 + k_2 r^4 + k_3 r^6) + p_1(r^2 + 2y^2) + 2p_2 xy \end{aligned} \quad (1)$$



Original Image



Radial Distortion



Tangential Distortion

Fig. 2 Distortion Types (Referenced from [4])

C. Camera Calibration

With the camera model and distortion model defined, one can conduct camera calibration. The "camera calibration problem" is the process of estimating unknown calibration parameters that ensure accurate 2D image to 3D space transformation. This problem is solved through the relationship of the "forward mapping problem" (1), the "reverse mapping problem" (2), and the "camera calibration problem" (3). Fig. 3 visually captures this relationship. "Predicting pixel coordinates (1), predicting line-of-sight directions in 3D space (2), and predicting camera parameters (3) are all complementary problems. One must know any two in order to estimate the third" (McKee, 2025). The reader is encouraged to reference [1] and [5] for a more in depth background of camera calibration, camera modeling and distortion modeling.

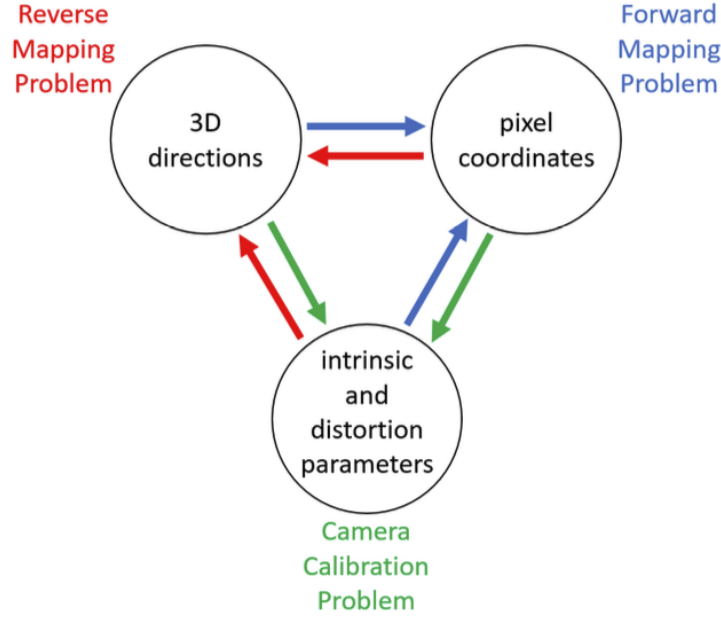


Fig. 3 Camera Calibration Relationship (Referenced from [1])

III. GIANT Utilization

This section aims to provide an overview of the steps taken to update spacecraft pose estimates using 2D images from a monocular camera aboard a spacecraft with the Goddard Imaging Analysis and Navigation Tool. In the results to follow, imagery from NASA's DAWN spacecraft mission to the asteroid Vesta is used to perform camera calibration, stellar OpNav, and relative OpNav. Important code snippets will be provided to show the ease of utilizing GIANT. Note that the pixel error is quantified as a residual error because there is no truth pose of the DAWN spacecraft at any point on its mission.

A. Camera Calibration

For camera calibration, each unique image is assigned a position, attitude, and epoch. Given these parameters, star identification algorithms can be ran to connect imaged stars to the corresponding stars referenced by the star catalog.

In Listing 1, the pinhole-Brown camera model is initialized, the DAWN camera is given a camera model and a set of images. Next, these resources are given to the calibration object.

```

# define pinhole Brown camera model
camera_model = BrownModel(fx=focal_length_pixels, fy=-focal_length_pixels, px=px, py=
    py, n_rows=nrows, n_cols=ncols, field_of_view=fov, estimation_parameters=
    estimation_parameters)

# camera object that contains our images
camera = DawnFCCamera(images=binned_images[0], model=camera_model, attitude_function=
    fc2_attitude)

# calibration object, containing the camera model and star catalogue
calib = Calibration(camera, star_id_kwargs={'catalogue': GIANTCatalogue()})

```

Listing 1 Camera Calibration Initialization

To obtain the most accurate camera calibration parameters it is necessary to refine each images attitude estimate prior to performing an estimate for the calibration parameters. Fig. 4 depicts the star identification results and the residual error between the star in the actual image and the corresponding identified catalog star's relative position projected to a pixel location before the attitude adjustment. Fig. 5 shows the same results after an attitude adjustment. It is shown that the star identification algorithm obtains more matches with decreased residual error after attitude refinement.

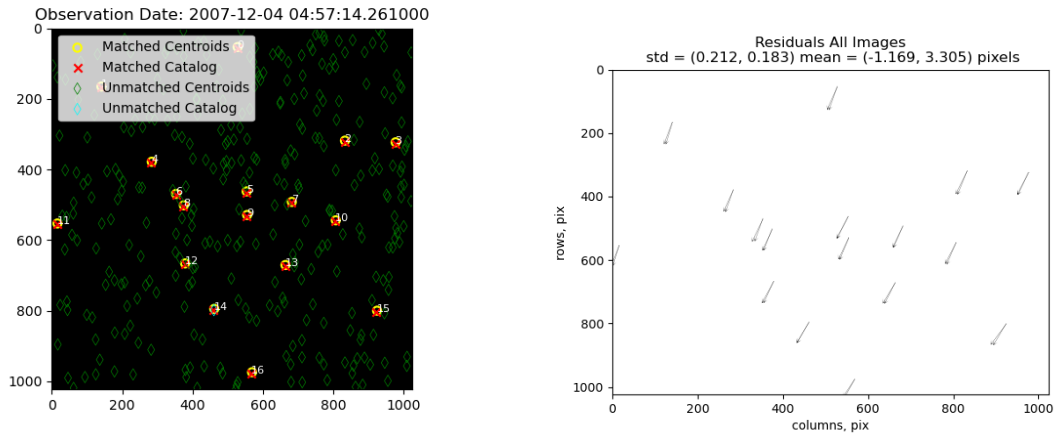


Fig. 4 Star ID and Pixel Residual Error Before Attitude Refinement

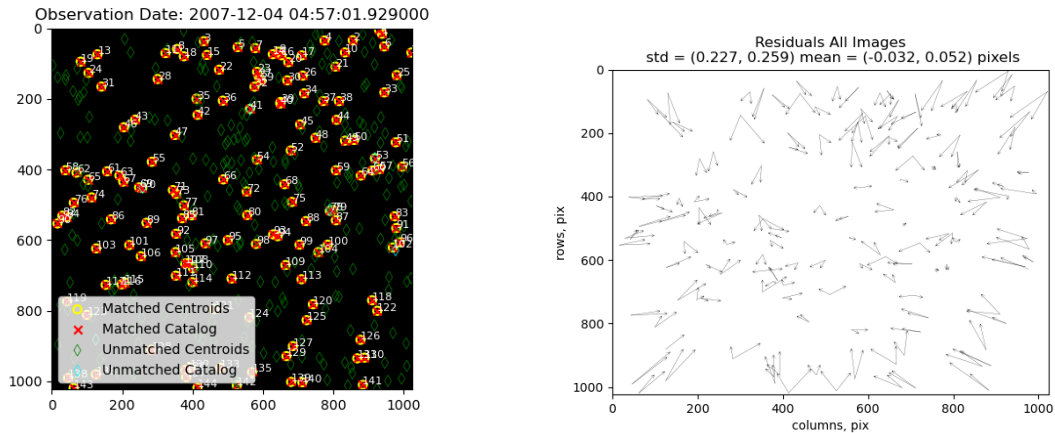


Fig. 5 Star ID and Pixel Residual Error After Attitude Refinement

With the attitude of each image refined, the camera calibration parameters can be estimated. Code representing star identification, attitude update, and the camera calibration processes is found in Listing 2. Only the x and y focal length in pixels (f_x, f_y), the corresponding r^2 distortion (k_1), and tangential distortion (p_1, p_2) calibration parameters were prescribed to be estimated. The resulting estimates of these calibration parameters are found in Table 1.

```
# now we can identify our stars and estimate an update to our attitude based on these
  matched star pairs
calib.id_stars()
calib.estimate_attitude()

# now we just want to identify stars, since we've already adjusted our attitude
calib.id_stars()

# estimate calibration parameters
calib.estimate_calibration()
```

Listing 2 Camera Calibration Execution

Table 1 Camera Intrinsic Parameters and Uncertainties

Parameter	Value	Uncertainty
f_x	1.072×10^4 pixels	4.7430×10^{-1}
f_y	-1.072×10^4 pixels	4.8290×10^{-1}
k_1	5.8390×10^{-2}	1.8880×10^{-2}
p_1	-5.684×10^{-4}	2.5420×10^{-4}
p_2	-5.139×10^{-4}	2.5620×10^{-4}

B. Optical Navigation

Once camera calibration is complete, the navigation engineer now has an accurate 2D image to 3D space transformation. Optical navigation can be separated into stellar and relative OpNav. Both goals result in different methods for generating pose correlated measurements from 2D images. Once a measurement is selected a method of estimating the selected measurement is embarked. Given a noisy measurement and a pose correlated estimate of that same measurement, a method of combining the measurement into the pose estimate can be employed. Both stellar and relative OpNav methods are provided to show the versatility of camera images.

1. Stellar Optical Navigation

Stellar navigation takes precisely the same form as camera calibration attitude refinement. For stellar OpNav the measurement is a 2D pixel location of an identified star. An estimate of that stars 2D pixel location must be obtained. To accomplish this end, a star identification algorithm is ran to connect imaged stars to stars in the star catalog. The current satellite position estimate and identified star position from the catalog can be used to compute the relative position vector between the camera and star. This relative position vector is then normalized to create a line-of-sight measurement, which is correlated to the attitude of the spacecraft. This 3D vector is then projected to a 2D pixel location with the camera and distortion model to compute a 2D pixel location estimate of the imaged star. Code for performing stellar OpNav is provided in Listing 3.

```

# load the camera model we are using
camera_model = load('dawn_camera_models.xml', 'FC2')
camera = DawnFCCamera(images=images, model=camera_model, psf=Gaussian(sigma_x=0.75,
    sigma_y=0.75, size=5),
    attitude_function=fc2_attitude)
# create stellar OpNav object
sopnav = StellarOpNav(camera, star_id_kwargs={'catalogue': GIANTCatalogue()})
# ID stars
sopnav.id_stars()
# perform attitude estimate
sopnav.estimate_attitude()

```

Listing 3 Stellar OpNav Execution

2. Relative Optical Navigation

The measurement created from an imaged body can take many forms. Some of which are center finding, ellipse matching, and limb matching. The results to follow use a center finding algorithm that assist the limb matching method to improve the spacecraft to Vesta relative position estimate. Generating a modeled measurement involves creating a fictitious "scene" where objects and light sources are created. The example provided contains a 3D model of Vesta and Sun light source modeled as a point. Given an image, there are algorithms that can find the centroid and pixels outlining the object of interest.

Detected pixel locations in the image are the measurements. Those same pixel locations projected into the fictitious "scene" by ray-tracing methods from our represent the model of the measurement. Code depicting this process is provided in Listing 4. Results from three epochs on approach are shown in Fig. 6. The residual error between the detected pixel and ray-traced detected point on Vesta to a pixel location is provided for the corresponding three epochs in Fig. 7. The orange can be thought of as the "pixel measurement" from the actual image of a point on Vesta and the blue as the "fictitious model prediction" of the same point on Vesta. At this point we can implement our noisy measurements into our state estimate with a process of choice.

```

# create 3D model of vesta
vesta_obj = SceneObject(vesta_shape, position_function=vesta_position,
    orientation_function=vesta_attitude, name='Vesta')

# model Sun point light source
sun_obj = SceneObject(Point([0, 0, 0]), position_function=sun_position,
    orientation_function=sun_orientation)

# Add Vesta and Sun to fictitious scene
opnav_scene = Scene(target_objs=[vesta_obj], light_obj=sun_obj)

# define the RelativeOpNav instance
relnav = RelativeOpNav(camera, opnav_scene,
    xcorr_kwargs=xcorr_kwargs,
    brdf=McEwenIllumination(),
    limb_matching_kwargs={'recenter': False},
    save_templates=True)

# incorporate measurements into pose estimate
relnav.auto_estimate()

```

Listing 4 Relative OpNav Execution

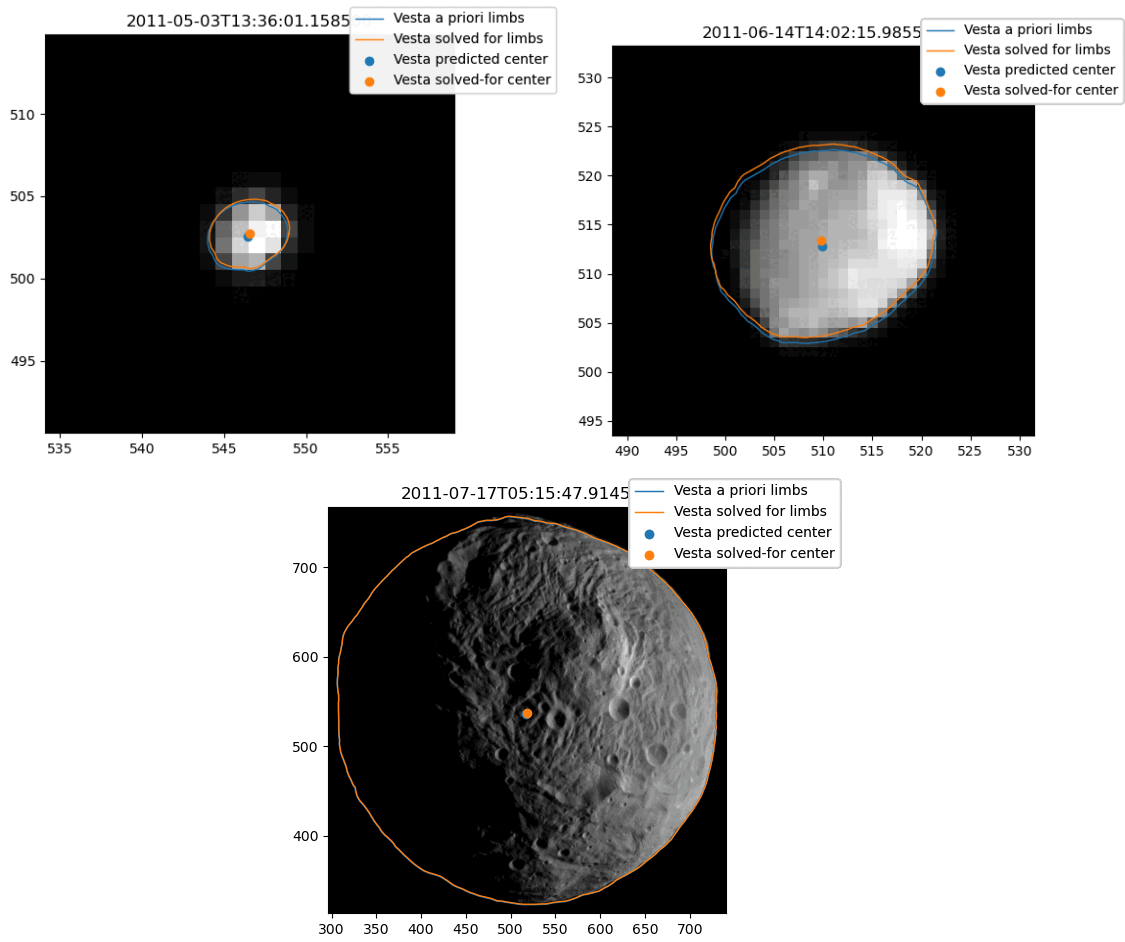


Fig. 6 Relative OpNav with Vesta Astroid

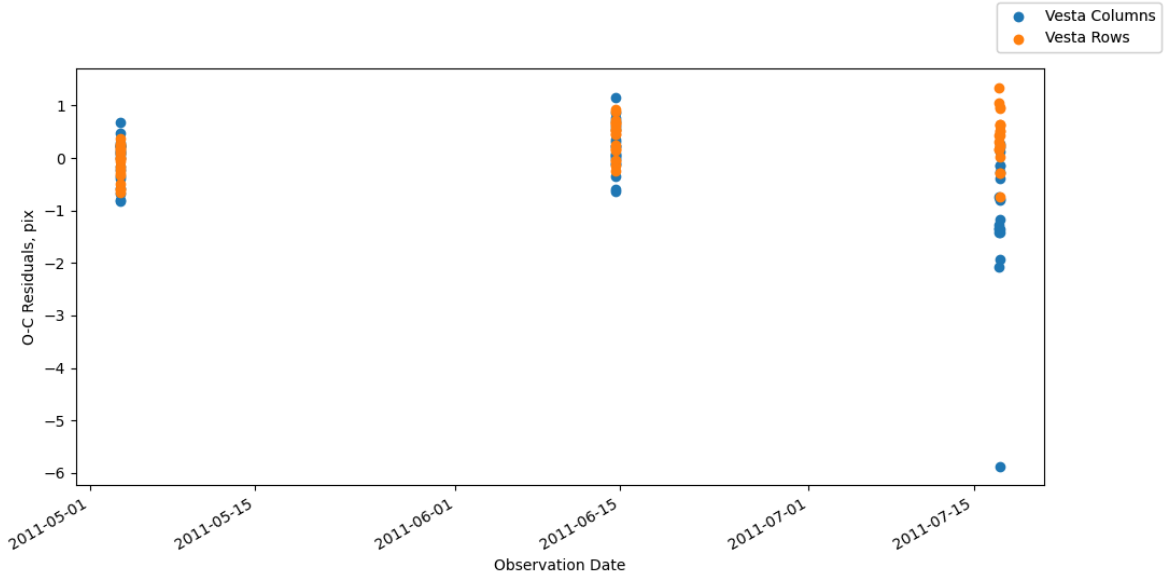


Fig. 7 Relative OpNav Residual Error

IV. Conclusion

In conclusion, a brief background of the process of estimating mathematically modeled calibration parameters that ensure accurate 2D image to 3D space transformation was described. With the 2D image to 3D space relationship established, the GIANT software tool was used to calculate calibration parameters and conduct optical navigation with images taken from NASA's DAWN mission to the asteroid Vesta. Short listings of code are provided to show the simplistic and effective utilization of GIANT without a complete understanding of underlying OpNav mathematics.

References

- [1] P. D. McKee, R. S. Lovelace, S. A. Lockhart, and J. B. Chong, "Camera Calibration and Alignment Metrology at Johnson Space Center's Electro-Optics Laboratory," 47th Annual AAS Guidance, Navigation and Control Conference, Breckenridge, CO, 2025, Paper No. AAS 25-141.
- [2] Liounis, A., "Goddard Image Analysis and Navigation Tool (GIANT)," NASA GitHub Repository, Accessed May 10, 2025. [Online]. Available: <https://github.com/nasa/giant>
- [3] Liounis, A., "GIANT: General Image Analysis and Navigation Toolbox Documentation," Accessed May 10, 2025. [Online]. Available: https://aliounis.github.io/giant_documentation/index.html
- [4] GeeksforGeeks, "CalibrateCamera() OpenCV in Python," Apr. 24, 2025. [Online]. Available: <https://www.geeksforgeeks.org/calibratecamera-opencv-in-python/>
- [5] Brown, D., "Decentering Distortion of Lenses," *Photogrammetric Engineering*, Vol. 32, No. 3, 1996, pp. 444–462.